

(a) Uniform I-JEPA

Uniform sampler

Smooth-L1

z_m ; stop-grad

$\hat{z}_m$

per-token LN
select T_m

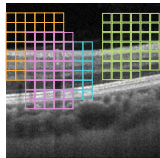
Index queries
 p_C, p_{T_m}
+ mask
tokens

Trainable g_ϕ
parallel targets

EMA teacher
 $f_{\bar{\theta}}$
no gradients

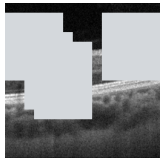
EMA weights

Online encoder
 f_θ
trainable



No guide

Sampler
 C, T_m



Visible C only

Target groups:



Evaluation: freeze EMA encoder $\rightarrow$ pool patches / B-scans $\rightarrow$ linear classifier

OCT: Kulyabin et al., OCTDL Fig. 1 (CC BY 4.0); gray = omitted patches, not zero input.

(b) Anatomy-guided I-JEPA

CENTROID example

Smooth-L1

z_m ; stop-grad

$\hat{z}_m$

per-token LN
select T_m

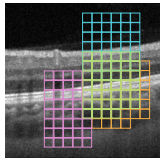
Index queries
 p_C, p_{T_m}
+ mask
tokens

Trainable g_ϕ
parallel targets

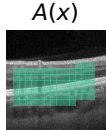
EMA teacher
 $f_{\bar{\theta}}$
no gradients

EMA weights

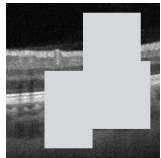
Online encoder
 f_θ
trainable



Full $x + T_m$ outlines



Sampler
 C, T_m



Visible C only

Target groups:



Evaluation: freeze EMA encoder $\rightarrow$ pool patches / B-scans $\rightarrow$ linear classifier

OCT: Kulyabin et al., OCTDL Fig. 1 (CC BY 4.0); gray = omitted patches, not zero input.